Tutorial 10
Counting and Probability

The Addition Rule

Recap:

Theorem 9.3.1 The Addition Rule

Suppose a finite set A equals the union of k distinct mutually disjoint subsets A_1, $A_2, \ldots, A_k$. Then

$$N(A) = N(A_1) + N(A_2) + \cdots + N(A_k).$$

Theorem 9.3.2 The Difference Rule

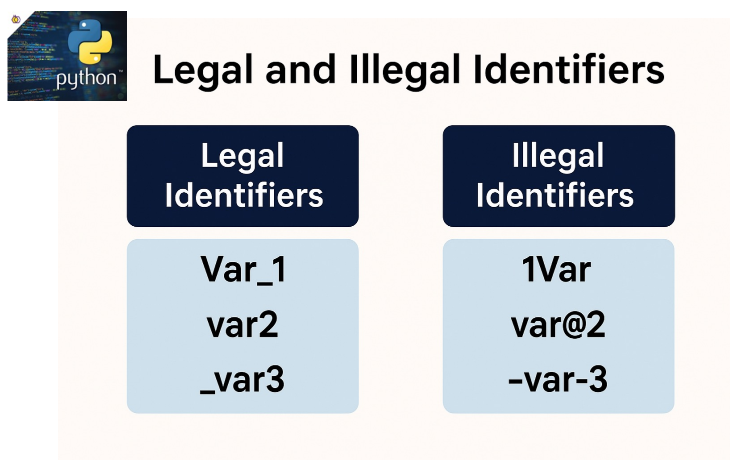
If A is a finite set and B is a subset of A, then

$$N(A - B) = N(A) - N(B).$$

Example 9.3.4-Number of Python identifiers of eight or Fewer Characters

Problem:

In the computer language Python, identifiers must start with one of 53 symbols: either one of the 52 letters of the upper- and lower-case Roman alphabet or an underscore (_). The initial character may stand alone, or it may be followed by any number of additional characters chosen from a set of 63 symbols: the 53 symbols allowed as an initial character plus the ten digits. Certain keywords, however, such as *and*, *if*, *print*, and so forth, are set aside and may not be used as identifiers. In one implementation of Python there are 31 such reserved keywords, none of which has more than eight characters. How many Python identifiers are there that are less than or equal to eight characters in length?



Example 9.3.4-Number of Python identifiers of eight or Fewer Characters

Solution The set of all Python identifiers with eight or fewer characters can be partitioned into eight subsets—identifiers of length 1, identifiers of length 2, and so on—as shown in Figure 9.3.4. The reserved words have various lengths (all less than or equal to 8), so the set of reserved words is shown overlapping the various subsets.

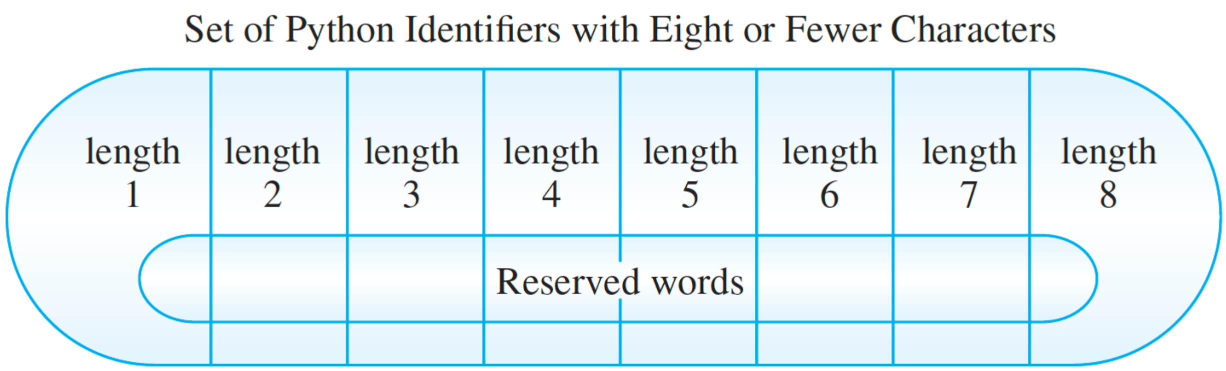
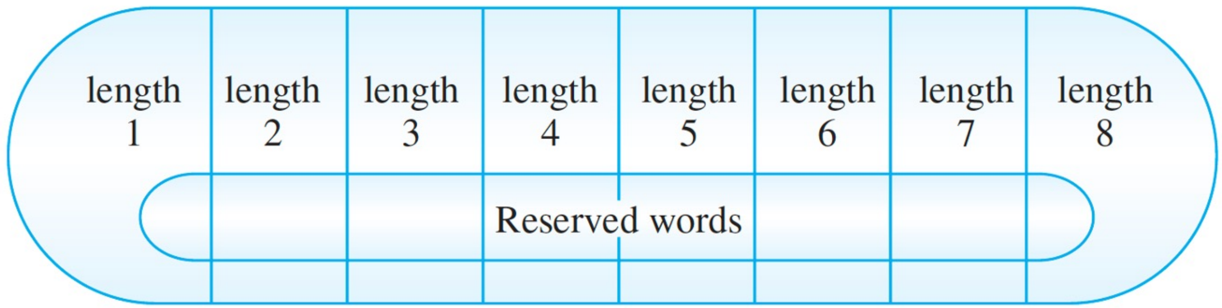


FIGURE 9.3.4

Example 9.3.4-Number of Python identifiers of eight or Fewer Characters

Set of Python Identifiers with Eight or Fewer Characters



Thus, by the addition rule, the number of potential Python identifiers with eight or fewer characters is

$$
\begin{aligned}
53 + 53\cdot 63 + 53\cdot 63^2 + 53\cdot 63^3 + 53\cdot 63^4 + 53\cdot 63^5 + 53\cdot 63^6 + 53\cdot 63^7 \\
&= 53\left(\frac{63^8 - 1}{63 - 1}\right) \quad \text{by Theorem 5.2.2} \\
&= 212{,}133{,}167{,}002{,}880.
\end{aligned}
$$

Now 31 of these potential identifiers are reserved, so by the difference rule, the actual number of Python identifiers with eight or fewer characters is

$$212{,}133{,}167{,}002{,}880 - 31 = 212{,}133{,}167{,}002{,}849. \quad \blacksquare$$

The Inclusion / Exclusion Rule

Recap:

Theorem 9.3.3 The Inclusion/Exclusion Rule for Two or Three Sets

If A, B, and C are any finite sets, then

$$N(A \cup B) = N(A) + N(B) - N(A \cap B)$$

and

$$N(A \cup B \cup C) = N(A) + N(B) + N(C) - N(A \cap B) - N(A \cap C) - N(B \cap C) + N(A \cap B \cap C).$$

Example 9.3.7-Counting the Number of elements in an intersection

Problem:

A professor in a discrete mathematics class passes out a form asking students to check all the mathematics and computer science courses they have recently taken. She found that, out of a total of 50 students in the class,

30 took precalculus;
18 took calculus;
26 took Python;
9 took both precalculus and calculus;
16 took both precalculus and Python;
8 took both calculus and Python;
47 took at least one of the three courses.

a. How many students did not take any of the three courses?

b. How many students took all three courses?

c. How many students took precalculus and calculus but not Python? How many students took precalculus but neither calculus nor Python?

Example 9.3.7-Counting the Number of elements in an intersection

Qa. How many students did not take any of the three courses?

Solution

a. By the difference rule, the number of students who did not take any of the three courses equals the number in the class minus the number who took at least one course. Thus the number of students who did not take any of the three courses is

$$50 - 47 = 3.$$

Example 9.3.7-Counting the Number of elements in an intersection

Qb. How many students took all three courses?

Solution

b. Let

$$P = \text{the set of students who took precalculus}$$
$$C = \text{the set of students who took calculus}$$
$$Y = \text{the set of students who took Python.}$$

Then, by the inclusion/exclusion rule,

$$N(P \cup C \cup Y) = N(P) + N(C) + N(Y) - N(P \cap C) - N(P \cap Y) - N(C \cap Y) + N(P \cap C \cap Y)$$

Substituting known values, we get

$$47 = 30 + 18 + 26 - 9 - 16 - 8 + N(P \cap C \cap Y).$$

Solving for $N(P \cap C \cap Y)$ gives

$$N(P \cap C \cap Y) = 6.$$

Hence there are six students who took all three courses. In general, if you know any seven of the eight terms in the inclusion/exclusion formula for three sets, you can solve for the eighth term.

Example 9.3.7-Counting the Number of elements in an intersection

Qc. How many students took precalculus and calculus but not Python? How many students took precalculus but neither calculus nor Python?

Solution

1. **Precalculus and Calculus but not Python**:
   - This is represented by the intersection $P \cap C$ minus the intersection of all three sets $P \cap C \cap Y$.
   - We already know that $|P \cap C| = 9$ and $|P \cap C \cap Y| = 6$.
   - So, the number of students who took Precalculus and Calculus but not Python is:

   $$|P \cap C \text{ but not } Y| = |P \cap C| - |P \cap C \cap Y| = 9 - 6 = 3$$

   Thus, **3 students** took Precalculus and Calculus but not Python.

2. **Precalculus but neither Calculus nor Python**:
   - This is the number of students who took only Precalculus. We calculate this as the total number of students who took Precalculus minus the students who took Precalculus and either Calculus or Python or both:

   $$\text{Only Precalculus} = |P| - (|P \cap C| + |P \cap Y| - |P \cap C \cap Y|)$$

   Substituting the values:

   $$\text{Only Precalculus} = 30 - (9 + 16 - 6) = 30 - 19 = 11$$

   Thus, **11 students** took Precalculus but neither Calculus nor Python.

r-Combinations with repetition Allowed

Recap:

Theorem 9.6.1

The number of r-combinations with repetition allowed (or multisets of size r) that can be selected from a set of n elements is

$$\binom{r+n-1}{r}.$$

This equals the number of ways r objects can be selected from n categories of objects with repetition allowed.

$$r = 15, n = 5$$

$$\times\ \times\ \times \mid \times\ \times\ \times\ \times\ \times\ \times\ \times \mid \mid \times\ \times\ \times \mid \times\ \times$$

A r-combination with repetition equals to a partition like this!

Example 9.6.5-The Number of integral Solutions of an equation

Problem:

How many solutions are there to the equation $x_1 + x_2 + x_3 + x_4 = 10$ if x_1, x_2, x_3, and x_4 are nonnegative integers?

Example 9.6.5-The Number of integral Solutions of an equation

Problem:

How many solutions are there to the equation $x_1 + x_2 + x_3 + x_4 = 10$ if x_1, x_2, x_3, and x_4 are nonnegative integers?

Solution

Think of the number 10 as divided into ten individual units and the variables x_1, x_2, x_3, and x_4 as four categories into which these units are placed. The number of units in each category x_i indicates the value of x_i in a solution of the equation. Each solution can, then, be represented by a string of three vertical bars (to separate the four categories) and ten crosses (to represent the ten individual units). For example, in the following table, the two crosses under x_1, five crosses under x_2, and three crosses under x_4 represent the solution $x_1 = 2$, $x_2 = 5$, $x_3 = 0$, and $x_4 = 3$.

Categories x_1		x_2		x_3		x_4	Solution to the Equation $x_1+x_2+x_3+x_4 = 10$
× ×	|	× × × × ×	|		|	× × ×	$x_1 = 2,\quad x_2 = 5,\quad x_3 = 0,\quad$ and $\quad x_4 = 3$
× × × ×	|	× × × × × ×	|		|		$x_1 = 4,\quad x_2 = 6,\quad x_3 = 0,\quad$ and $\quad x_4 = 0$

$$\binom{10+3}{10} = \binom{13}{10} = \frac{13!}{10!(13-10)!} = \frac{13\cdot 12\cdot 11\cdot \cancel{10!}}{\cancel{10!}\cdot 3\cdot 2\cdot 1} = 286.$$

Example 9.6.6-Additional Constraints on the Number of Solutions

Problem:

How many integer solutions are there to the equation $x_1 + x_2 + x_3 + x_4 = 10$ if each $x_i \geq 1$?

Example 9.6.6-Additional Constraints on the Number of Solutions

Problem:

How many integer solutions are there to the equation $x_1 + x_2 + x_3 + x_4 = 10$ if each $x_i \geq 1$?

Solution

Transform the problem to:

$$(x_1 - 1) + (x_2 - 1) + (x_3 - 1) + (x_4 - 1) = 10 - 4$$

Both sides minus 4

=> $y_1 + y_2 + y_3 + y_4 = 6$

Each $y_i \geq 0$

$$\binom{6+3}{6} = \binom{9}{6} = \frac{9!}{6!(9-6)!} = \frac{9 \cdot 8 \cdot 7 \cdot \cancel{6!}}{\cancel{6!} \cdot 3 \cdot 2 \cdot 1} = 84.$$

Pile of Coins

Problem:

A large pile of coins consists of pennies, nickels, dimes, and quarters.

(1) How many different collections of 30 coins can be chosen if there are at least 30 of each kind of coin?

Pile of Coins

(1) How many different collections of 30 coins can be chosen if there are at least 30 of each kind of coin?

Solution

🔷 (1) At least 30 of each coin type

Here, there are **no upper bounds** on the variables. We just want the number of integer solutions to:

$$p + n + d + q = 30, \quad p, n, d, q \geq 0$$

This is a classic **stars and bars** problem:

$$\text{Number of solutions} = \binom{30 + 4 - 1}{4 - 1} = \binom{33}{3} = \boxed{5456}$$

Pile of Coins

Problem:

A large pile of coins consists of pennies, nickels, dimes, and quarters.

(2) If the pile contains only 15 quarters but at least 30 of each other kind of coin, how many collections of 30 coins can be chosen?

Pile of Coins

(2) If the pile contains only 15 quarters but at least 30 of each other kind of coin, how many collections of 30 coins can be chosen?

Solution

◆ **(2) Only 15 quarters available:** $q \leq 15$

Same setup as before, but now with constraint:

$$p + n + d + q = 30, \quad 0 \leq q \leq 15$$

We want to **exclude** solutions where $q > 15$.

Let's compute the **total unrestricted solutions** as before: $\binom{33}{3} = 5456$

Now subtract the number of "bad" solutions where $q > 15$. Let's change variable:

Let $q' = q - 16 \geq 0 \Rightarrow q = q' + 16$

So the equation becomes:

$$p + n + d + (q' + 16) = 30 \Rightarrow p + n + d + q' = 14$$

Number of such solutions:

$$\binom{14 + 4 - 1}{4 - 1} = \binom{17}{3} = 680$$

Thus:

$$\text{Valid solutions} = \boxed{5456 - 680 = 4776}$$

Pile of Coins

Problem:

A large pile of coins consists of pennies, nickels, dimes, and quarters.

(3) If the pile contains only 20 dimes but at least 30 of each other kind of coin, how many collections of 30 coins can be chosen?

Pile of Coins

(3) If the pile contains only 20 dimes but at least 30 of each other kind of coin, how many collections of 30 coins can be chosen?

Solution

◆ (3) Only 20 dimes available: $d \leq 20$

Similar logic:

- Total (unrestricted): $\binom{33}{3} = 5456$

To subtract "bad" cases where $d > 20$, let $d' = d - 21 \geq 0$

So equation becomes:

$$p + n + (d' + 21) + q = 30 \Rightarrow p + n + d' + q = 9 \Rightarrow \binom{12}{3} = 220$$

So:

$$\text{Valid solutions} = \boxed{5456 - 220 = 5236}$$

Pile of Coins

Problem:

A large pile of coins consists of pennies, nickels, dimes, and quarters.

(4) If the pile contains only 15 quarters and only 20 dimes but at least 30 of each other kind of coin, how many collections of 30 coins can be chosen?

Pile of Coins

(4) If the pile contains only 15 quarters and only 20 dimes but at least 30 of each other kind of coin, how many collections of 30 coins can be chosen?

Solution

◆ **(4) Only 15 quarters and only 20 dimes available:** $q \leq 15, d \leq 20$

We apply **inclusion-exclusion**:

- Total (no restrictions): $T = \binom{33}{3} = 5456$
- Subtract:
  - A: Bad solutions with $q > 15$: already computed as 680
  - B: Bad solutions with $d > 20$: already computed as 220
- Add back the **overlap**: both $q > 15$ **and** $d > 20$

For the overlap, define:

- $q' = q - 16 \geq 0$
- $d' = d - 21 \geq 0$

Then:

$$p + n + d' + q' = 30 - 16 - 21 = -7 \Rightarrow \text{No solutions} \Rightarrow \binom{-4}{3} = 0$$

So:

$$\text{Valid solutions} = T - A - B + (A \cap B) = 5456 - 680 - 220 + 0 = \boxed{4556}$$

Remark: Deciding Which Formula to Use

The following table summarizes which formula to use when choosing k elements from n in different scenarios:

	Order Matters	**Order Does Not Matter**
Repetition Is Allowed	n^k	$\binom{k+n-1}{k}$
Repetition Is Not Allowed	$P(n, k)$	$\binom{n}{k}$

Probability Axioms

Recap:

Probability of a General Union of Two Events

If S is any sample space and A and B are any events in S, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \qquad 9.8.2$$

Extension:

$$\begin{aligned} P(A_1 \cup A_2 \cup A_3) = P(A_1) &+ P(A_2) + P(A_3) \\ &- P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) \\ &+ P(A_1 \cap A_2 \cap A_3) \end{aligned}$$

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i) - \sum_{1 \le i < j \le n} P(A_i \cap A_j) + \sum_{1 \le i < j < k \le n} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1} P(A_1 \cap A_2 \cap \cdots \cap A_n)$$

Probability Axioms

Problem:

A person writes four letters to four friends and then randomly puts them into four envelopes that are already addressed, with one letter in each envelope. What are the probabilities of the following scenarios?

(1) All four letters are put into the correct envelopes;
(2) All four letters are put into the wrong envelopes.

Probability Axioms

Solution

◆ **(1) $P(A)$: Probability that all 4 letters are correctly placed**

The solution computes:

$$P(A) = P(A_1 A_2 A_3 A_4)$$

That is, the probability that each letter A_i goes into its correct envelope.

This is done by multiplying the conditional probabilities:

$$P(A) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1A_2) \cdot P(A_4|A_1A_2A_3)$$

- First letter: 1 correct choice out of 4 → $\frac{1}{4}$
- Second: 1 correct out of 3 → $\frac{1}{3}$
- Third: 1 correct out of 2 → $\frac{1}{2}$
- Fourth: only 1 left, and it must match → 1

So:

$$P(A) = \frac{1}{4} \cdot \frac{1}{3} \cdot \frac{1}{2} \cdot 1 = \frac{1}{24}$$

Probability Axioms

Solution

◆ **(2) $P(B)$: Probability that all 4 letters are in the wrong envelopes**

Let $B = A_1^c A_2^c A_3^c A_4^c$, meaning **every letter is placed incorrectly**.

Instead of calculating this directly, the solution uses:

$$P(B) = 1 - P(\text{at least one letter is correct}) = 1 - P(A_1 \cup A_2 \cup A_3 \cup A_4)$$

This is the union of events where each letter goes to the correct envelope. To compute this, they use the **inclusion-exclusion principle**:

$$P(A_1 \cup A_2 \cup A_3 \cup A_4) = \sum P(A_i) - \sum P(A_i A_j) + \sum P(A_i A_j A_k) - P(A_1 A_2 A_3 A_4)$$

Substituting the values:

- $P(A_i) = \frac{1}{4}$, and there are 4 such terms
- $P(A_i A_j) = \frac{1}{12}$, and there are $\binom{4}{2} = 6$ such pairs
- $P(A_i A_j A_k) = \frac{1}{24}$, and there are $\binom{4}{3} = 4$ such triplets
- $P(A_1 A_2 A_3 A_4) = \frac{1}{24}$, only one such case

So:

$$P(A_1 \cup A_2 \cup A_3 \cup A_4) = 4 \cdot \frac{1}{4} - 6 \cdot \frac{1}{12} + 4 \cdot \frac{1}{24} - \frac{1}{24} = 1 - \frac{1}{2} + \frac{1}{6} - \frac{1}{24} = \frac{15}{24}$$

Therefore:

$$P(B) = 1 - \frac{15}{24} = \frac{9}{24} = \boxed{\frac{3}{8}}$$

Thank You!